

Program : Diploma in Textile Technology	
Course Code : 3067	Course Title: Textile wet Processing I Lab
Semester : 3	Credits: 1.5
Course Category: Program core	
Periods per week: : 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart practical knowledge on preparatory and colouration process
- To prepare the recipes for the complete wet processing treatments
- To analyze the effectiveness of the process.

Course Prerequisites:

Topic	Course code	Course name	Semester
Fundamentals of Textile Fibres		Textile Fibres	2

Course Outcomes :

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Identify and Interpret the given fibres	5	Applying
CO2	Execute the preparatory process of grey cotton & viscose	10	Applying
CO3	Demonstrate the colouring of cotton & viscose material using water soluble dyes and after treatment.	16	Applying
CO4	Demonstrate the dyeing of cotton/ viscose material using water-insoluble dyes and analyse the fastness properties	11	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3						
CO3	3						
CO4		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Identify and Interpret the given fibres		
M1.01	Interpret the given fibres using burning test – Cotton, Wool, Silk. Determination of composition of fibres and their blend ratio in the given sample of (P/C) yarn / fabric.	2	Applying
M1.02	Interpret the given fibres using burning test – Viscose, Polyester, Nylon, Acetate & Acrylic. Determination of composition of fibres and their blend ratio in the given sample of (P/V) yarn / fabric.	3	Applying
CO2	Execute the preparatory process of grey cotton & viscose		
M2.01	Desizing of cotton fabric with enzyme.	2	Applying
M2.02	Scouring of Desized cotton material with NaOH	2	Applying
M2.03	Bleaching of scoured cotton with hydrogen peroxide.	3	Applying
M2.04	Mercerization of cotton fabric/yarn hank under slack conditions and to test the changes	3	Applying
	Lab Exam - I	1.5	
CO3	Demonstrate the colouring of cotton & viscose material using water soluble dyes and after treatment		
M3.01	Dyeing of cotton/viscose material with Direct dyes	4	Applying
M3.02	Test the Fastness properties of dyed material and Apply after treatments to direct dyed materials for Fastness to light, wash, and perspiration	4	Applying

M3.03	Dyeing of cotton material with Cold brand reactive dyes	4	Applying
M3.04	Dyeing of cotton material with Hot brand reactive dyes	4	Applying
CO4	Demonstrate the dyeing of cotton/ viscose material using water-insoluble dyes and analyse the fastness properties		
M4.01	Dyeing of cotton material with Sulphur dyes	2	Applying
M4.02	Dyeing of cotton/viscose material with Vat dyes and analyse the fastness properties	5	Applying
M4.03	Dyeing of cotton/viscose material with Naphthols	4	Applying
	Lab Exam - II	1.5	

The students should keep record for wet processing with samples and submit certified & bonafide record at the time of practical examination.

Text / Reference:

T/R	Book Title/Author
T1	V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan publishers Ahmedabad.
R2	R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose fibres
R3	E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres
R4	Srivastava – Recent process in Textile Bleaching and Dyeing
R5	R RChakravarthi – A Glimpse of the chemical technology of Textile fibres
R6	Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/116/102/116102016/
2	https://swayam.gov.in/
3	http://eacharya.inflibnet.ac.in/vidya-mitra/
4	www.youtube.com